

Intro to AI Agents

AI Agents

An artificial intelligence (AI) agent is a software program that can interact with its environment, collect data, and use the data to perform self-determined tasks to meet predetermined goals.

Humans set goals, but an AI agent independently chooses the best actions it needs to perform to achieve those goals

AI Agent - Example

A contact center AI agent that wants to resolves customer queries.

The agent will automatically ask the customer different questions, look up information in internal documents, and respond with a solution.

Based on the customer responses, it determines if it can resolve the query itself or pass it on to a human.

AI Agent Types – Simple Reflex

Act based solely on current perceptions using condition-action rules.

These agents respond directly to stimuli without considering past experiences or potential future states.

They operate on basic "if-then" logic: if a specific condition is detected, execute a corresponding action.

AI Agent Types – Simple Reflex

Key traits

- No memory of past states
- No model of how the world works
- Purely reactive behavior
- Function best in fully observable environments

Examples

- Thermostats that turn heating on/off based on current temperature
- Traffic light control systems that change signals based on fixed timing
- Basic automatic doors that open when motion is detected

AI Agent Types – Model-based reflex

Maintain an internal representation of the world, allowing them to track aspects of the environment they cannot directly observe.

This internal model helps them make more informed decisions by considering how the world evolves and how their actions affect it.

AI Agent Types – Model-based reflex

Key traits

- Track the world's state over time
- Infer unobserved aspects of current states
- Function effectively in partially observable environments
- Still primarily reactive, but with contextual awareness

Examples

- Robot vacuum cleaners that map rooms and track cleaned areas
- Smart home security systems that monitor multiple entry points
- Automated inventory systems that track stock levels

AI Agent Types – Goal-based

Plan their actions with a specific objective in mind.

Unlike reflex agents that respond to immediate stimuli, goal-based agents evaluate how different action sequences might lead toward their defined goal, selecting the path that appears most promising.

AI Agent Types – Goal-based

Key traits

- Employ search and planning mechanisms
- Evaluate actions based on their contribution toward goal achievement
- Consider future states and outcomes
- May explore multiple possible routes to a goal

Examples

- Navigation apps finding the fastest route to a destination
- Chess-playing programs planning several moves ahead
- Automated scheduling systems optimizing resource allocation

AI Agent Types – Utility-based

Extend goal-based thinking by evaluating actions based on how well they maximize a utility function—essentially a measure of "happiness" or "satisfaction."

This approach allows them to make nuanced trade-offs between competing goals or uncertain outcomes.

AI Agent Types – Utility-based

Key traits

- Balance multiple, sometimes conflicting objectives
- Handle probabilistic and uncertain environments
- Evaluate actions based on expected utility
- Make rational decisions under constraints

Examples

- Self-driving cars balancing speed, safety, and fuel efficiency
- Stock trading algorithms weighing risk against potential returns
- Resource management systems in cloud computing environments

AI Agent Types – Learning

Improve their performance over time based on experience.

They modify their behavior by observing the consequences of their actions, adjusting their internal models and decision-making approaches to achieve better outcomes in future interactions.

AI Agent Types – Learning

Key traits

- Adapt to changing environments
- Improve performance with experience
- Contain both a performance element and a learning element
- Generate new knowledge rather than simply applying existing rules

Examples

- Recommendation systems that improve suggestions based on user feedback
- Chatbots that adapt their responses to better meet user needs
- Game-playing AI that develops new strategies through practice

AI Agent Types – Learning

Key traits

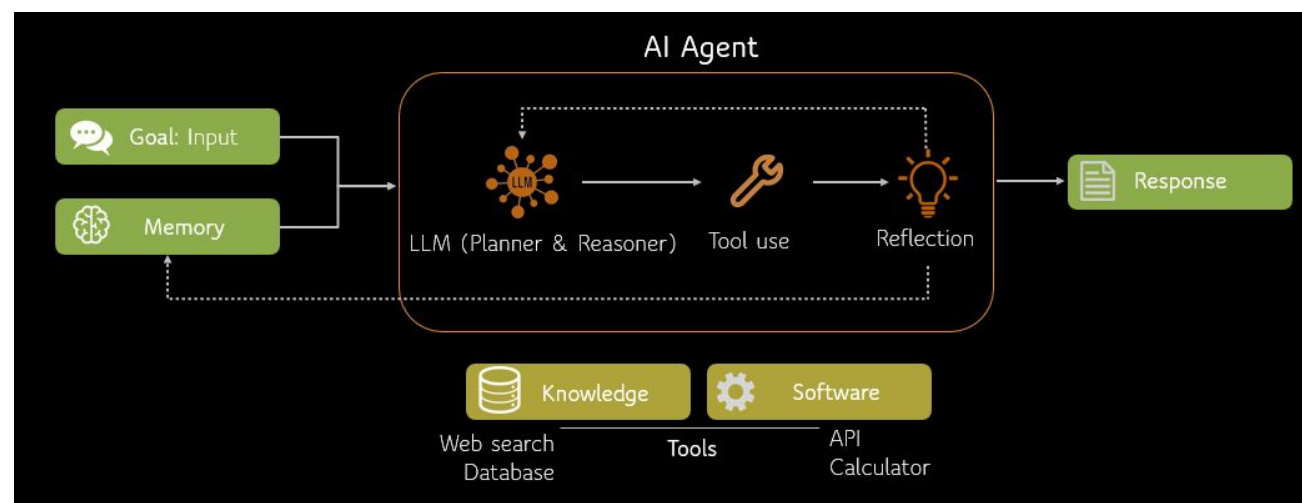
- Adapt to changing environments
- Improve performance with experience
- Contain both a performance element and a learning element
- Generate new knowledge rather than simply applying existing rules

Examples

- Recommendation systems that improve suggestions based on user feedback
- Chatbots that adapt their responses to better meet user needs
- Game-playing AI that develops new strategies through practice

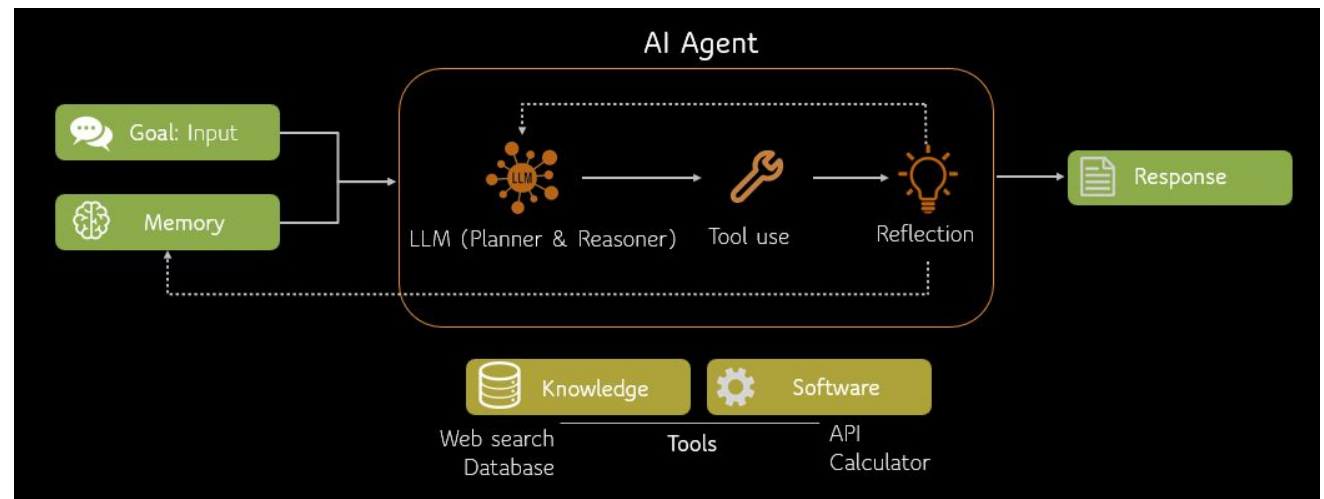
Agents Components - Goal

Goal: the objective that the agent is trying to achieve and that is usually set by user (e.g. “Summarize today’s news and send a short email,” or “Find the top 5 candidates for this job posting.”).



Agents Components - Planning

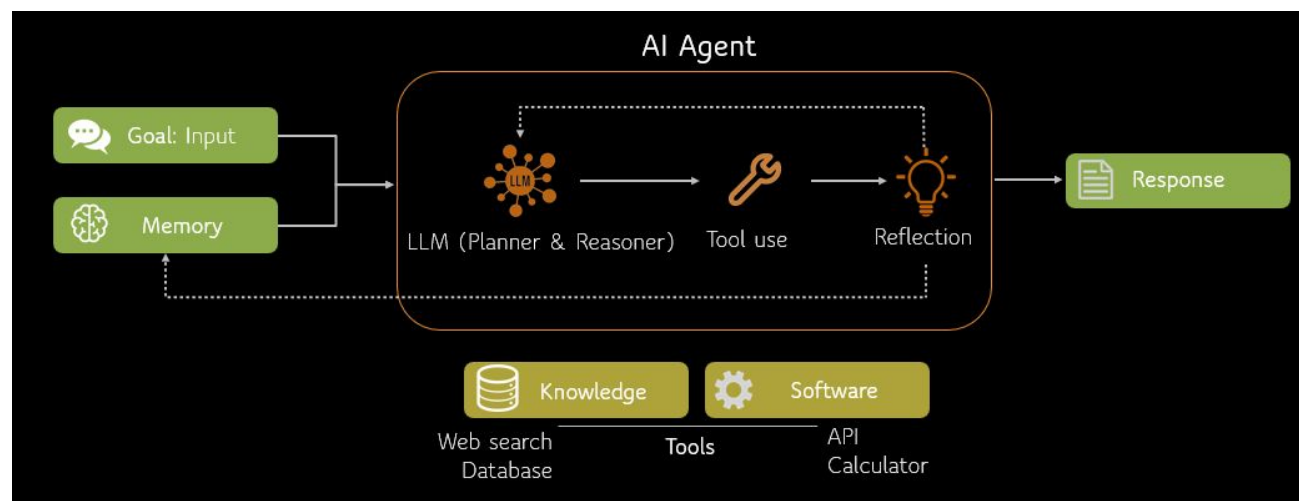
Planning: the logic that decides on the next steps. It can be either rule-based, or (usually) utilizes an LLM to reason about which step or tool is needed to move toward the goal.



Agents Components - Tools

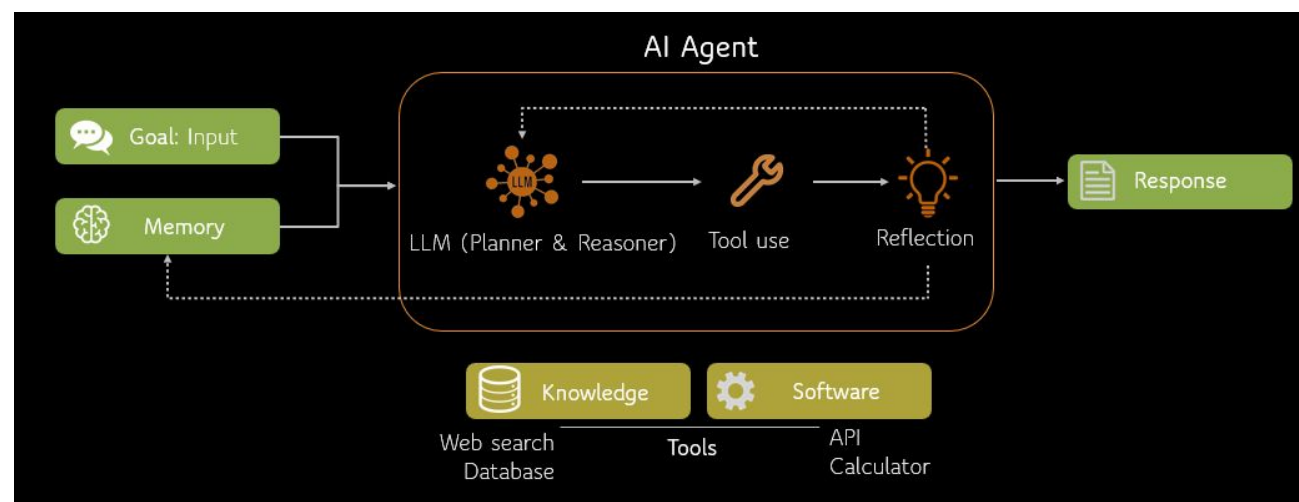
Tool use: tools are external functions that the agent can call to interact with the environment (APIs, databases, file systems, search engines, etc.).

The LLM suggests a structured function call to access real actions and up-to-date information.



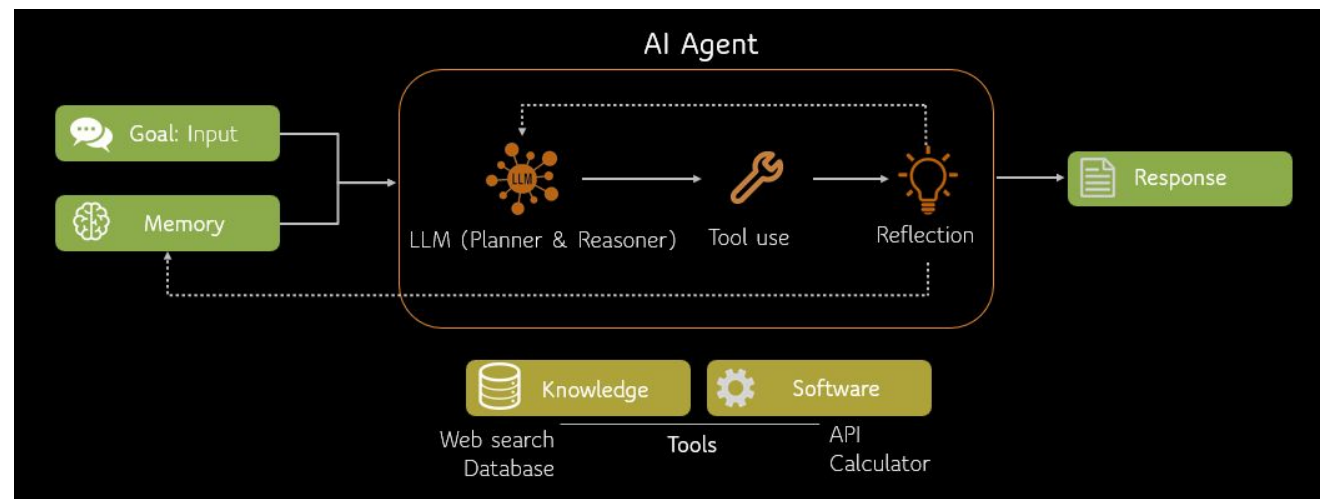
Agents Components - Memory

Memory: mechanisms for the agent to remember information across steps and sessions. **Episodic memory** is short-term context from previous turns. **Semantic memory** includes long-term facts or knowledge (such as vector embeddings of past conversations or documents)



Agents Components - Reflection

- **Reflection and Feedback loop:** the agent's ability to evaluate its output and retry if needed. Self-reflection helps to avoid compounding errors, iterate and ensure that the quality of the output is satisfactory. It is usually done via prompts like "Was that result satisfactory?" or "What should be done next?".



Agents Flow

- Receive a goal,
- Decide on the next action,
- Call a tool (if required),
- Store results in memory,
- Evaluate the quality of results,
- Repeat until the goal is achieved.

